

5. Acquisition A: noise-plate series

Series A measures depth noise on the flat plate T2. Its pose list holds the nine ladder stations at the five field positions and the two legacy depths, 700 and 1000 mm (LEGACY_METRIC_DEPTHS_MM), at the center only, with 100 frames at each pose (FRAMES_PER_NOISE_STATION), plus a tilt sub-series. The plan tool shuffles the pose order with a logged seed, so that slow drift cannot masquerade as a Z dependence.

1. Mount T2 and do the mount check of Step 4.8. The plate must cover the analysis region at every pose. Where the requested field position would carry part of the plate out of the image, the plan tool pulls the pose inward along its field direction until the plate stays inside the image with a margin of 8 px (BOUNDARY_BAND_HALF_WIDTH_PX), records the fraction of the requested offset it achieved in the pose list, and the A results report that fraction per station. At the near stations this happens for the off-axis positions.
2. Execute the pose list in its order; the seed of the shuffle is logged with the plan.
3. Before the first pose, capture a drift sentinel: T2 centered at the reference station, 800 mm (Z_REFERENCE_MM), 30 frames (SENTINEL_FRAMES). A sentinel is repeated every 60 minutes (DRIFT_SENTINEL_INTERVAL_MIN) and after the last pose of each series. During B, C, and D the sentinel uses the front plane of whatever target is mounted, centered at the reference station, and the first sentinel after each mount is that target's reference; the T2 sentinels bracket each series, so no re-mount is needed for a sentinel.
4. At each station: move, wait ROBOT_SETTLE_TIME_S, then capture FRAMES_PER_NOISE_STATION consecutive frames. Log the read-back robot pose, the sensor temperature, and timestamps; the air temperature comes from the loggers of Step 4.1.
5. Tilt sub-series. At the center of the reduced stations (every fourth station of the ladder: 400, 800, 1600 mm), tilt T2 about V, then about H, through 0, 15, 30, and 45° (TILT_ANGLES_DEG), with 50 frames per pose (FRAMES_PER_TILT_POSE). A tilt is allowed only where the plate's near edge stays at or beyond Z_MIN, that is $Z - (\text{half the plate width}) \times \sin(\text{tilt}) \geq Z_{\text{MIN}}$; the plan tool skips the other poses and lists them. With the 400 mm plate this rules out every tilt at 400 mm, so the sub-series runs at 800 and 1600 mm. Incidence angle changes both the per-pixel noise and the fill rate.
6. Repeat-mount check. Dismount T2, re-mount it, and repeat the Z_REFERENCE_MM center station. The difference shows how much of the bias comes from re-mounting the target.

7. Jitter-span check. Before any B, C, or D capture, run the autocorrelation quick-look of Section 10, Step 7 on one center station. If the correlation length exceeds PHASE_JITTER_SPAN_PX, raise the span (Section 2) before continuing.
8. If Step 4.2 calls for it, repeat Steps 1–6 with the sensor's filters off; this repeat is outside the Section 9 budget.